INCH POUND
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MILITARY SPECIFICATION

REFRIGERATORS; FROZEN FOOD CABINETS; AND COMBINATION REFRIGERATORS-FROZEN FOOD CABINETS, MECHANICALLY REFRIGERATED, COMMERCIAL, SELF-CONTAINED, NAVAL SHIPBOARD

This interim amendment is approved for use by all departments and Agencies of the Department of Defense.

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1.1: Delete and substitute the following:

"1.1 Scope. This document covers commercial mechanical refrigerators, thaw refrigerators, frozen food cabinets, combination units and modular constructed refrigerators and frozen food cabinets and combination units including explosion-proof equipment for Naval shipboard use."

1.2, under "Classification". Add in the proper sequence,

"Type IV - Thaw Refrigerator".

1.2, Add line 8, "Model I - Modular Construction".

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2.1, under "STANDARDS/MILITARY" add the following:

"MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristies, Measurement of.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities."

AMSC N/A FSC 4110

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3.2.4 line 4: Last sentence, after "acceptable": Add "For modular constructed units, in the assembly process only those types of fastening methods can be used which would allow the refrigerated cabinet to be disassembled at a future date without the use of special tools or methods which might damage or disfigure the equipment. No permanent fastening means of the sub-components to one another is acceptable, such as welding, brazing, soldering, or the use of epoxies or other adhesives. The use of screws or rivets as a primary method of attaching components is not acceptable. To insure a permanent attachment between the metal sheeting of the interior and exterior to the insulation a bonding agent is required. The preferred method of achieving the adhesion is by applying a mastic material to the metal components prior to insulating. Mastic shall be a synthetic rubber base specifically designed to adhere urethane and metal with excellent water resistance properties."

Add the following new paragraphs:

"3.2.4.1 Panel fasteners. Cam lock type field assembly fasteners that utilize a friction driven hook with 1/2" eccentric cam pull up to capture, draw up and lock together with a pin assembly in an adjacent panel."

"3.2.4.2 Fastener assembly. A two piece housing comprised of 16 guage cold rolled steel, flanged on the rear for bearing on the structure's perimeter member, may be secured by nails through provided holes or integral spines. Contained within the housing is an assembly of a high pressure die cast cam, a spring steel friction clutch, both within a full hard cold rolled steel stamped hook. Activation is by use of a 5/16 hex wrench inserted through a receiving hole in the cam. Rotation of the wrench rotates the hook cam and spring assembly within the housing, bringing out the hook. Upon the hook contacting the pin in the adjacent panel, the friction clutch releases, causing the eccentric cam to draw minimum tensile load of the assembly 1500 lbs."

"3.2.4.3 Pin assembly. A two piece housing formed of 16 gauge cold rolled steel with a double headed steel 1/4" diameter pin riveted between. Flanged to bear on the panel structure perimeter member."

"3.2.4.4 Access caps. The caps used to seal the panel fastener wrench access hole, when installed should be water tight and have flared edges of not greater than 135 degrees. The access caps shall be of an impervious vinyl or similiar material that is approved and listed by the National Sanitation Foundation (NSF)."

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3.4.1, line 5, after "retention": Add "A requirement of the modular equipment is that it be capable of being installed in the location it is to be used with a minimum of 3 inches of space in excess of the finished height and width of the cabinet. The total installation of the refrigerated cabinet is to be accomplished from the front of the unit without access to the exterior sides, back and top."

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Add as new paragraphs:

"3.4.6 Vibration. Refrigeration equipment shall withstand ship vibration Controls, valves, switches, moving parts and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage (see 4.4.10).

3.4.7 Electromagnetic compatibility. When electromagnetic compatibility is required (see 6.2), the units shall be equipped for electromagnetic compatibility in accordance with the requirements of MIL-STD-461, class A4 for surface ships and class A5 for submarines. The equipment shall meet the emission and susceptibility requirements for CE01, CE03, and RE02 of MIL-STD-461 (see 4.4.11).

3.4.8 Human factors criteria. Human factors engineering criteria, principles, and practices in accordance with MIL-STD-1472, shall be incorporated into the design of the refrigerating equipment. The equipment shall be designed such that all maintenance and operation shall permit safe mmd efficient performances by the 5th percentile female to the 95th percentile male in accordance with MIL-STD-1472. All controls, switches, valves and gauges shall be selected and integrated into the design of the ranges so as to meet the application requirements of MIL-STD-1472 and they shall be clearly and appropriately labeled to identify function. The clearance or free area required around an item shall permit an individual with applicable 5th to 95th percentile body dimensions and physical capabilities in accordance with MIL-STD-1472, to safely operate maintain, remove, or replace that item. When establishing accessibility requirements, both physcial and visual access must be provided, along with access for any tools, test equipment, and replacement parts needed. When inspecting for defects and performing tests, the equipment shall adhere to the human factors engineering considerations listed herein."

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Add the following new paragraphs:

"3.5.1.2 Panel construction. For modular constructed cabinets the insulated panels, which form the cabinet, shall join together in a tongue-and-groove method. The tongue must fit into the groove a minimum of 1/2 inch and the fit shall be such that the panel can not be moved laterally 1/16 of an inch once the panels are in place but not fastened. The insulated cabinet panels shall be attached to one another by means of cam lock panel fasteners. The panel fasteners shall be the primary method of attachment and shall not be spaced more than 18 inches apart. The fasteners shall have an individual holding capacity of 1200 lbs. The seams created as a result of ajoining panels shall be designed in such a way as to be self sealing. The seal shall be such that it does not permit the entry or leaking of water. Gasketing between the panels shall not be used in forming a seal between two panels. Lumber, steel, of any other conductive material shall not be used in forming the perimeter edges of the panel. Instead the perimeter edges shall be formed of polyurethane foam with an in place density of ten pounds per cubic foot. To insure that the perimeter edges are structurally significant, they shall be a minimum cross section of 1 1/2" high x 2 1/2" wide. The polyurethane edge shall be molded and not shaped so that it will have a resilient finish and be impervious to liquids and odors. The refrigerated cabinet must be structurally integral, meaning that it must possess, when assembled, all qualities of conventional shipboard refrigerated cabinets.

Add as new paragraph:

"3.5.2.4.2 Modular refrigeration systems. "The refrigeration system shall be furnished fully assembled, and operational. It will have been previously charged, tested and all controls properly adjusted. The system shall have the capability of being partially disassembled and reassembled, at the time of installation in the vessel, without the need of an additional refrigerant charge or control adjustments. The condendensing unit shall be able to be removed from the condensing unit compartment for ease in moving the cabinet base section."

Add as new paragraph:

"3.5.2.8 Modular refrigerant couplings. "The refrigeration lines shall be connected by using self sealing refrigerant couplings. The couplings are to be specifically designed for refrigeration applications and shall be used in sets (liquid and suction lines). There shall be a minimum of two sets in a system. Larger systems may require additional sets. The refrigerant couplings are to be a type which would allow for future system opening at the coupling without its failure."

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The maximum operating pressure (PSI) shall conform to the following:

Maximum Operating Pressure (PSI) Complete Coupling Assembly (Connected)		3000 PSI
Minimum Burst Pressure (PSI) (Connected)		9000 PSI
Maximum Operating Pressure (PSI) (Disconnected)	Male Half	2500 PSI
	Female Half	500 PSI
Air Inclusion (c.c. Max)		.10
Fluid Loss (c.c. Max)		.05
Vacuum Test (in./hg)		28
Maximum Rated Flow (GPM)		2

The coupling shall also be able to be replaced by unscrewing them from the respective line.

3.5.3 line 3: after "grounded": Add "In modular units the electrical materials as well as methods of construction and installation shall be in full compliance with the National Electrical Manufacturers Association (NEMA) standards."

3.5.3. line 5: after "grounds within the equipment.": Add "In modular units all electrical wiring feeding the interior must be concealed in the walls or run through the exterior back through a protected raceway. No wires shall be exposed in the interior or on any other area of the refrigerated cabinet except within the refrigeration compartment."

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Add as new paragraph:

"3.5.3.3.3 Modular units electrical circuitry. "The electrical wiring circuitry shall be in place, integral to the system, and fully operational at the time of equipment delivery. It also must have the capability to be partially disassembled, at the time of installation on the vessel, and reconnected without requiring any special tools. Additionally, there shall not be the need to make any connections inside of electrical junction boxes or with methods requiring handling of exposed bare wire ends. Electrical connections shall be by means of polarized molded pin and socket devices. Each field electrical connector shall be compatible with only one other connector in the system. This is to disallow any possibility of wrong connections."

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3.5.4 line 19, after "requirements": Add "In modular units the interior corners shall be radiused as defined by the National Sanitation Foundation under Standard 7. Such corners shall not be formed when two panels are joined and shall not have a seam within one inch of the center of the radius."

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3.5.5.2 line 7, after "drain": Add " The drain line from the refrigerator evaporator coil shall run through a penetration in the rear wall and enclosed in a raceway along the exterior back and into the refrigeration compartment. The design of the drain line shall be such as to allow it to remain attached to the rear wall during disassembly and installation of the cabinet. All that shall be required for installation is to couple the drain line to the bottom fitting on the evaporator coil."

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4.2.3, Table VII, add the following:

Characteristic	Requirement	Test
"Vibration	3.4.6	4.4.10
Electromagnetic compatibility	3.4.6	4.4.11"

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Add as new paragraphs:

"4.4.10 Vibration. The refrigeration equipment, under normal operating conditions, shall be tested in accordance with MIL-STD-167-1, type I equipment. The equipment shall be secured to the test machine in the same manner that it will be secured on shipboard (see 3.5.1). Failure of the refrigeration equipment to perform its function during and after testing, or meeting requirements of 3.4.6 shall constitute failure of this test.

4.4.11 Electromagnetic interference. The unit shall be tested in accordance with test methods CE01, CE03, and RE02 of MIL-STD-462. Nonconformance with the requirements specified shall constitute failure of the test,"

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Add as new paragraph:

"6.1.1 Modular constructed units. Cabinets sizes up to 20 cubic-feet shall be integral construction unless otherwise specified (see 6.2.m). Cabinets over 20 cubic-feet shall be modular constructions. Various sizes for each type, model and style (see tables I, II, III, IV, V, VI, and VII). The modular concept, by design, is intended to circumvent the requirement to cut access openings or enlarge passageways in vessels to accommodate conventionally constructed shipboard refrigerators, freezers and combination units."

6.2 Add in the proper sequence:

"m. When cabinets under 20 cubic-feet shall be modular construction."

Preparing activity:
Navy - SA
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